IPL Match Simulation Web Application

JAVA FULL STACK MINI PROJECT

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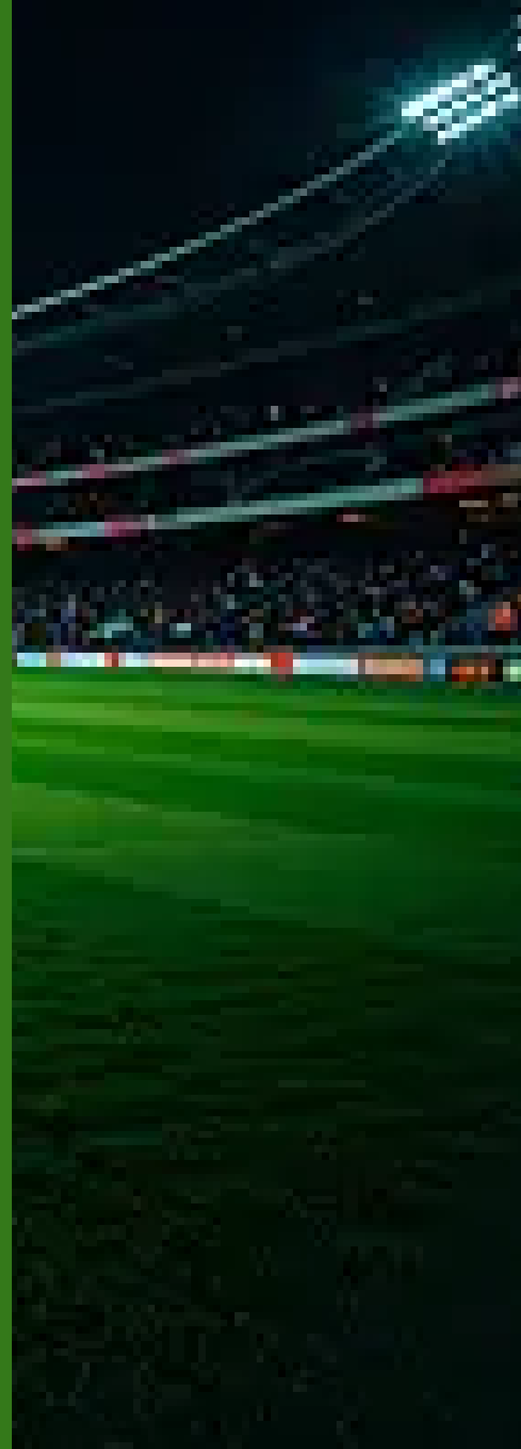
Problem Statement & Objective

Problem Statement:

- Existing platforms only provide static scores or live streaming
- No interactive system for simulating cricket matches
- Lack of real-time engagement and dynamic match experience

Objective:

- Develop a web-based IPL match simulation system
- Provide ball-by-ball live updates and commentary
- Maintain player statistics and match records
- Enable admin control for teams, players, and scheduling



- **Architecture Layers:**

- Presentation Layer ➛ JSP, HTML, CSS, JS
- Application Layer ➛ Servlets + Simulation Engine
- Data Layer ➛ MySQL Database

- **Workflow:**

- User selects a match from homepage
- Match initialized by server
- User clicks “Next Ball”
- Request sent to backend (Servlet)
- Simulation engine generates outcome
- Data stored in MySQL database
- UI updates instantly using AJAX

Key Features

- Live match simulation (ball-by-ball)
- Real-time score updates with commentary
- Player profiles with detailed statistics
- Orange Cap & Purple Cap leaderboards
- Complete match scorecards
- Admin panel (manage teams, players, matches)
- Auto-play feature for continuous simulation
- Fantasy Point Arena

Technologies & Conclusion

Technologies Used:

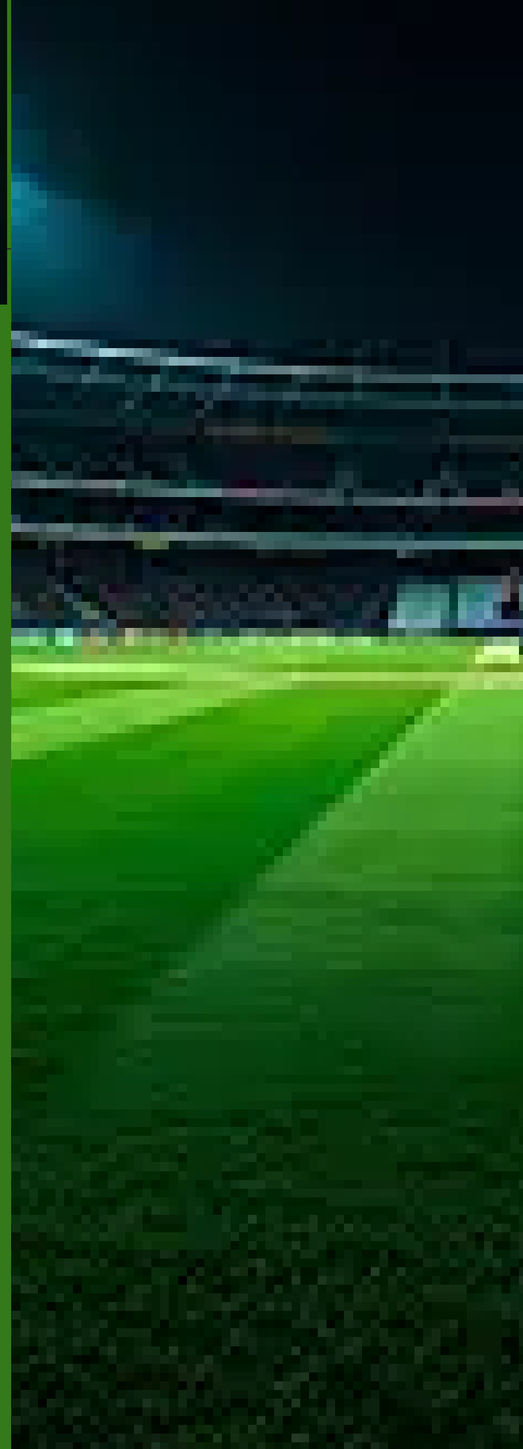
- Frontend: HTML, CSS, JavaScript (AJAX)
- Backend: Java Servlets, JSP (Apache Tomcat)
- Database: MySQL (JDBC)
- Build Tool: Maven

Conclusion:

- Successfully built a real-time IPL simulation system
- Demonstrates full-stack development without frameworks
- Provides interactive and engaging user experience
- Can be enhanced with features like real NRR, better AI logic, and security improvements

FUTURE SCOPE

- Sports analytics for team strategy and performance evaluation in Indian Premier League
- Match outcome prediction based on player statistics and past data
- Used in fantasy sports platforms like Dream11 for scoring and team selection
- Training tool to simulate different match situations and strategies
- Basis for cricket video games and sports simulation systems



THANK YOU